

		$\lambda = 0.50 \times 10^{-9} \text{ m}$
18	B	Using $x = \frac{\lambda D}{a}$, since D is constant, $x \propto \frac{\lambda}{a}$,

No.	Answer	Solution
		Option A: $\frac{\lambda}{a} = \frac{700 \times 10^{-9}}{4.0 \times 10^{-3}} = 176 \times 10^{-6}$ Option B: $\frac{\lambda}{a} = \frac{20 \times 10^{-3}}{50 \times 10^{-3}} = 0.4$ Option C: $\frac{\lambda}{a} = \frac{450 \times 10^{-9}}{2.0 \times 10^{-3}} = 225 \times 10^{-6}$ Option D: $\frac{\lambda}{a} = \frac{10 \times 10^{-3}}{200 \times 10^{-3}} = 0.05$ For maximum x, the magnitude of $\frac{\lambda}{a}$ is maximum, i.e. 0.4.
10.	A	Time proton spends between the parallel plates